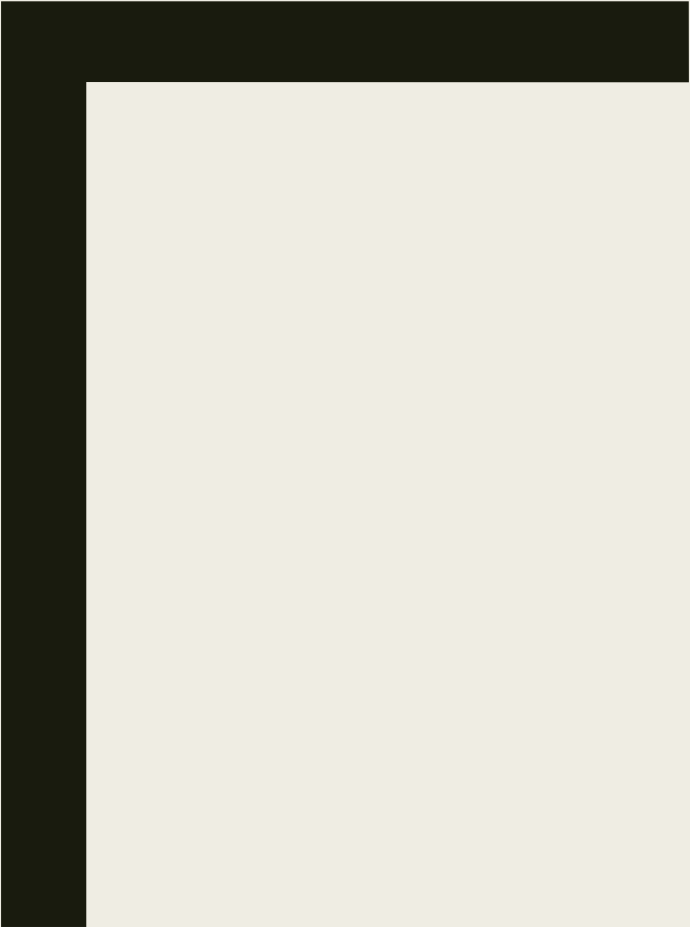


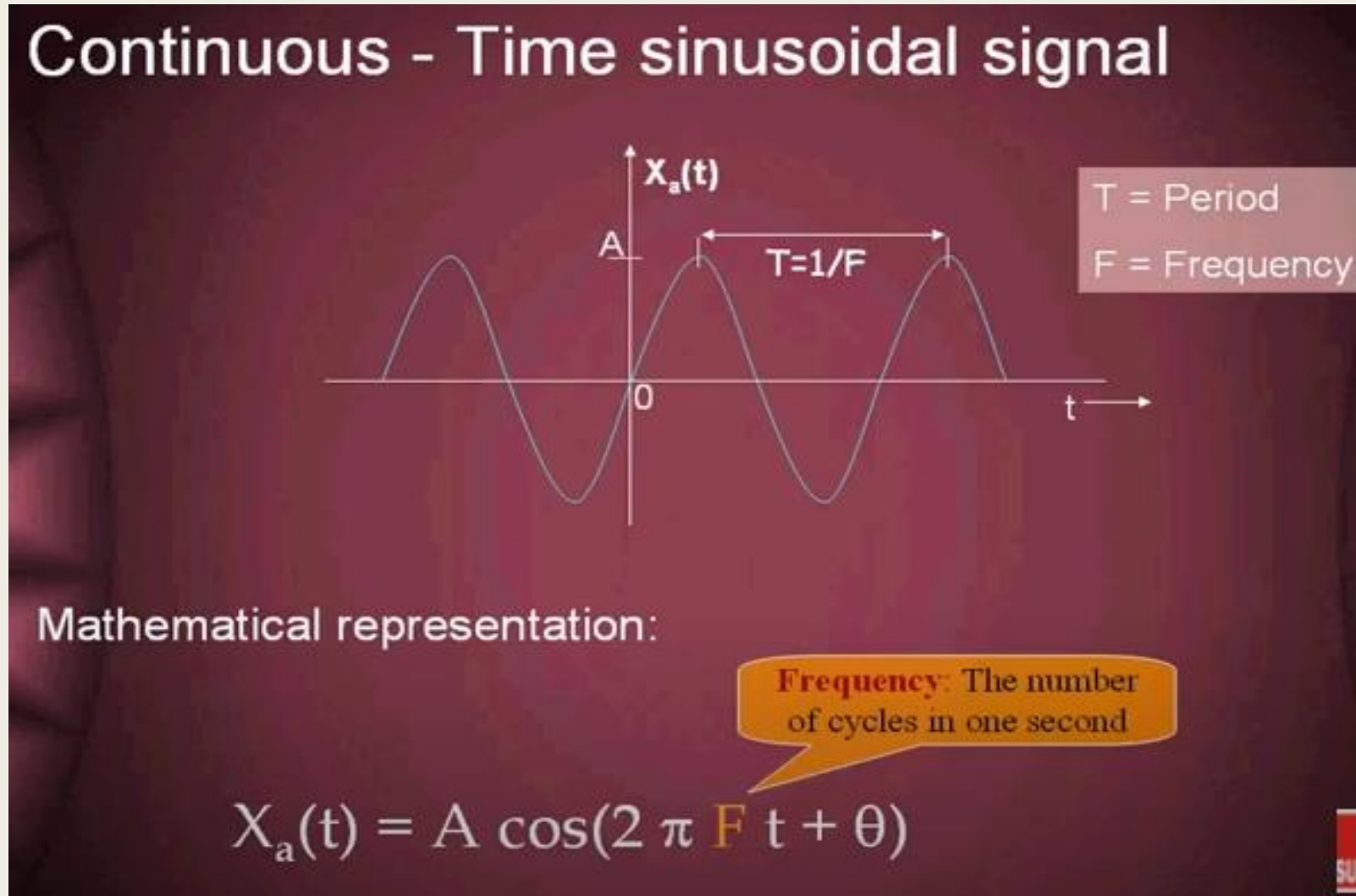


LEC 4: CHAPTER 1

Dr. Arsla Khan

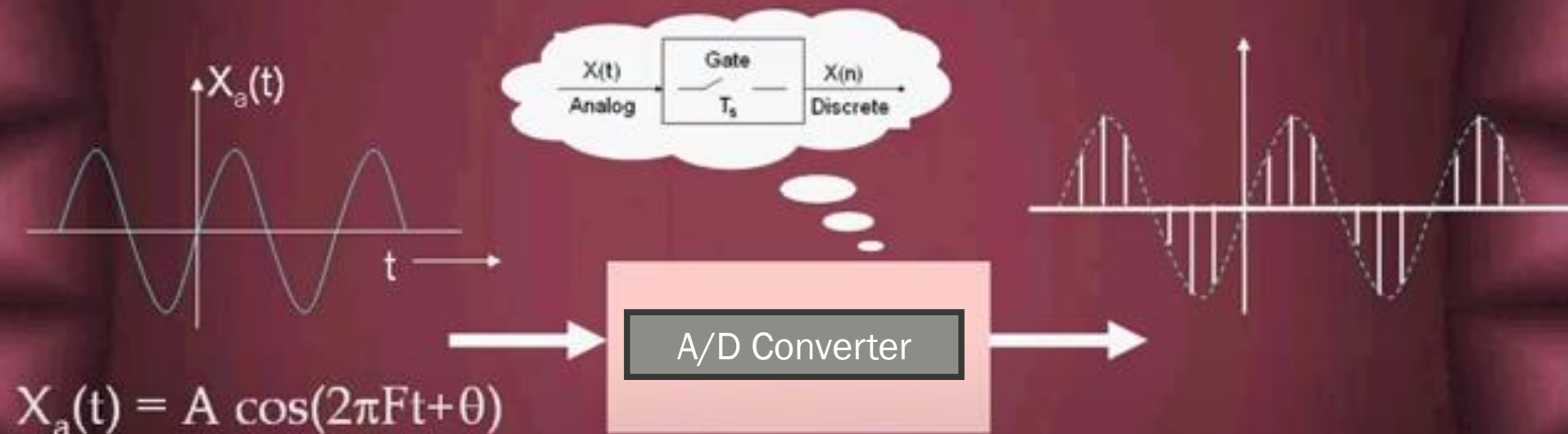


Concept of Frequency



Discrete - Time sinusoidal signal

“ The sampling process ”



The concept of Digital Frequency

$$X_a(t) = A \cos(2\pi F t + \theta) \quad t = nT_s$$

$$X(n) = A \cos(2\pi F n T_s + \theta)$$

$$T_s = \frac{1}{F_s}$$

$$X(n) = A \cos(2\pi F n \frac{1}{F_s} + \theta)$$

Digital
Frequency

$$f = \frac{F}{F_s}$$

$$\text{Digital Frequency} = \frac{\text{Analog Frequency}}{\text{Sampling Frequency}}$$

Understanding Digital Frequency

$$\text{Digital frequency} = f = \frac{F}{F_s}$$

- ◆ Normalized Frequency
- ◆ Discrete Time Frequency

Digital Frequency is defined as the frequency with which the samples of a discrete sinusoid occur.

Lets look at its dimension

$$f = \frac{\text{Cycles / sec}}{\text{Samples / sec}} = \frac{\text{Cycles}}{\text{Sample}}$$

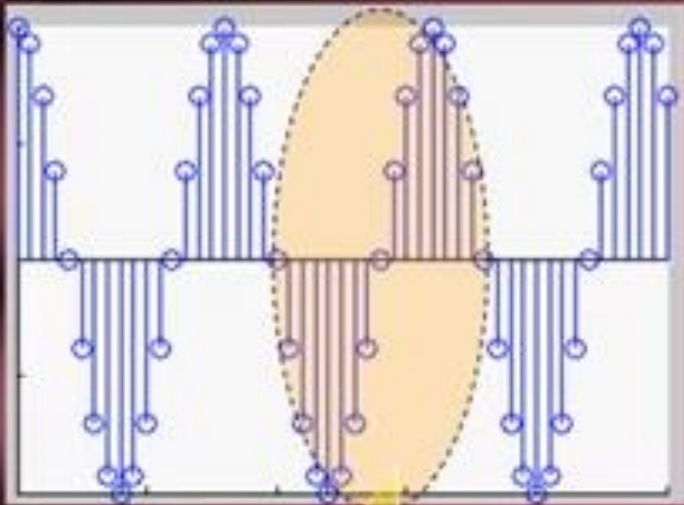
Digital frequency has the dimensions of **number of cycles per sample**

Example:

$$X(n) = \cos(2\pi fn)$$

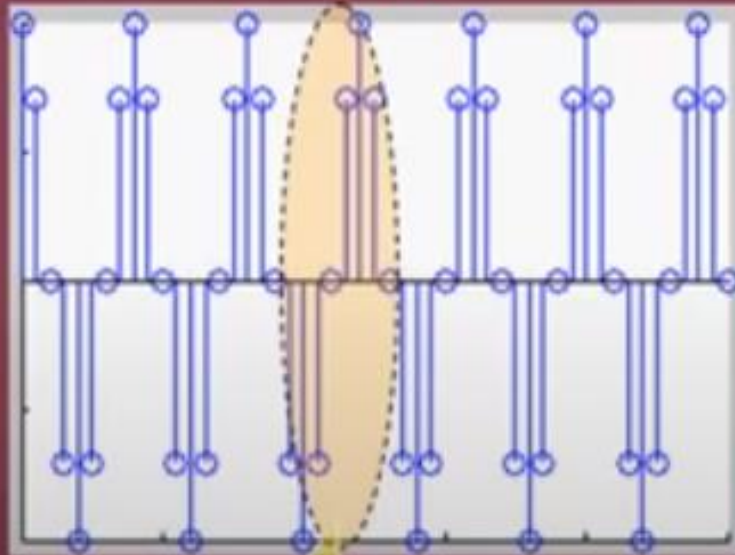
$$f = 1/16, 1/8, 1/4$$

$$f = 1/16$$



16 samples
in a cycle

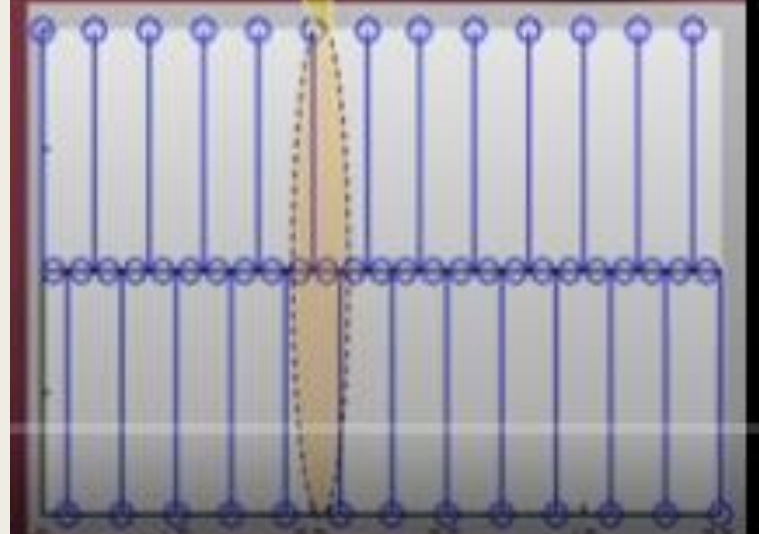
$$f = 1/8$$



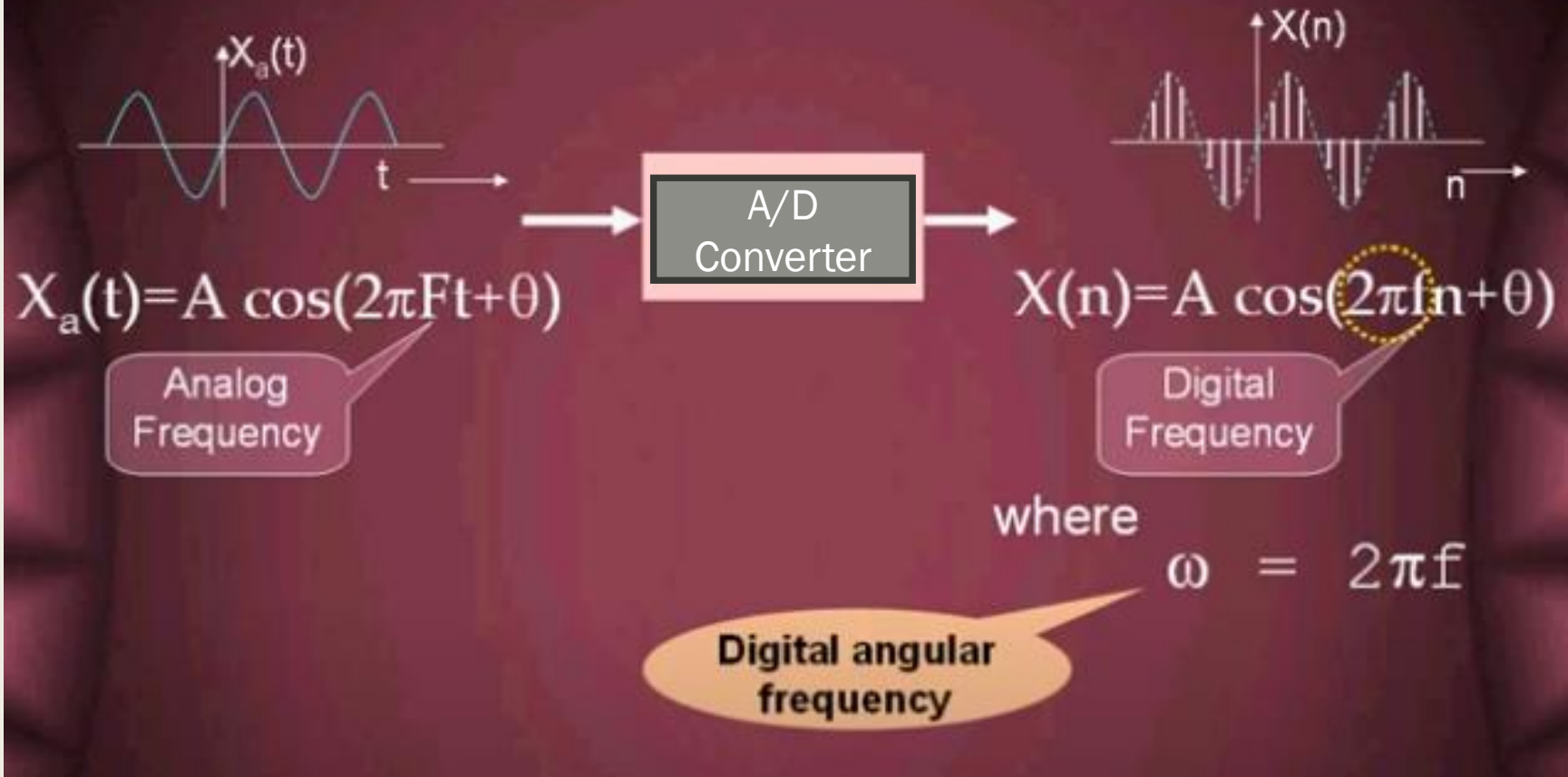
8 samples
in a cycle

4 samples
in a cycle

$$f = 1/4$$



Ideal analog/digital converter



- Mathematical expression for discrete time sinusoidal signal is

$$X[n] = A \cos(\omega n + \theta)$$

Property of discrete time sinusoid

$$X(n) = A \cos(\omega n + \theta)$$

$$\omega + 2\pi$$

$$X'(n) = A \cos((\omega + 2\pi)n + \theta)$$

$$= A \cos(\underbrace{(\omega n + \theta)}_A + \underbrace{2\pi n}_B)$$

$$= A \cos(\omega n + \theta)$$

$$= X(n)$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(\omega n + \theta) \cos(2\pi n) - \sin(\omega n + \theta) \sin(2\pi n)$$

one

zero

- Discrete time sinusoids whose frequencies are separated by an integer multiple of 2π are identical

Digital frequency range

$$0 \leq \omega \leq 2\pi$$



$$0 \leq 2\pi f \leq 2\pi$$



$$0 \leq f \leq 1$$

$$-\pi \leq \omega \leq \pi$$



$$-\pi \leq 2\pi f \leq \pi$$



$$-1/2 \leq f \leq 1/2$$

Frequency range of discrete time signals is finite

Fundamental range: $0 \leq \omega \leq 2\pi$

or

$$-\pi \leq \omega \leq +\pi$$

Sampling

$$X_1(t) = \cos(2\pi F_1 t)$$

$$F_1 = 100\text{Hz}$$

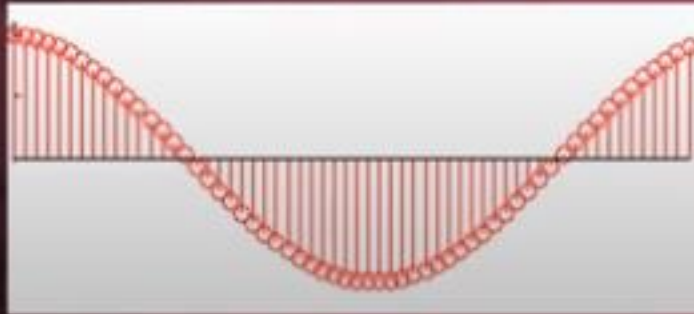
$$X_2(t) = \cos(2\pi F_2 t)$$

$$F_2 = 900\text{Hz}$$

$$\text{Sampling rate } F_s = 6500 \text{ Hz}$$

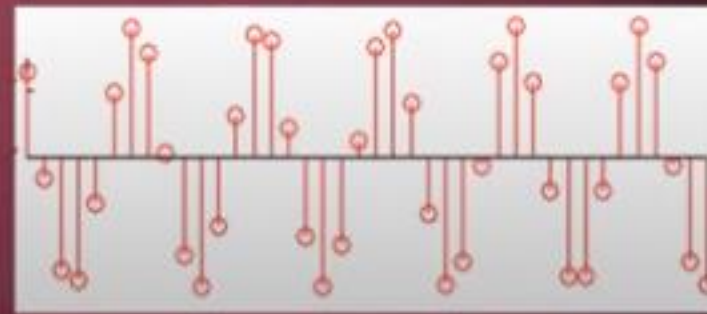
$$f_1 = F_1 / F_s = 1/65$$

$$X_1(n) = \cos(2\pi f_1 n)$$



$$f_2 = F_2 / F_s = 1/7$$

$$X_2(n) = \cos(2\pi f_2 n)$$



An accurate digital representation is possible only if sampling frequency is sufficiently high

$$X_1(t) = \cos(2\pi F_1 t)$$

$$F_1 = 100\text{Hz}$$

$$X_2(t) = \cos(2\pi F_2 t)$$

$$F_2 = 900\text{Hz}$$

With $F_s = 1000\text{ Hz}$

$$f_1 = F_1/F_s = 1/10$$

$$X_1(n) = \cos(2\pi f_1 n)$$

$$f_2 = F_2/F_s = 9/10$$

$$X_2(n) = \cos(2\pi f_2 n)$$



Mathematical proof

$$X_2(n) = \cos(2\pi f_2 n)$$

$$f_2 = 9/10$$

$$X_2(n) = \cos(2\pi [9/10] n)$$

$$X_2(n) = \cos(2\pi n [1 - 1/10])$$

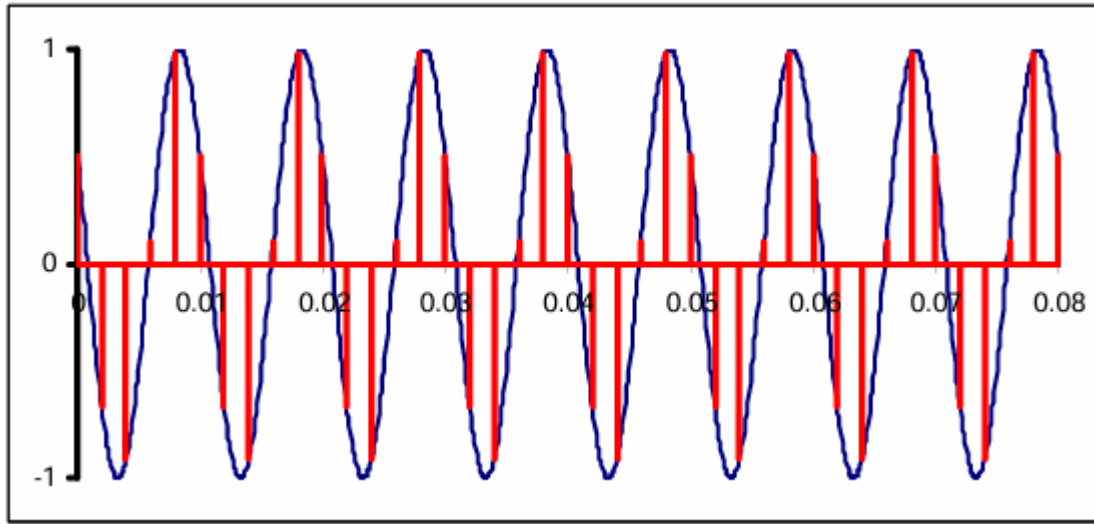
Simplifying using $\cos(A+B)$ identity

$$f_1$$

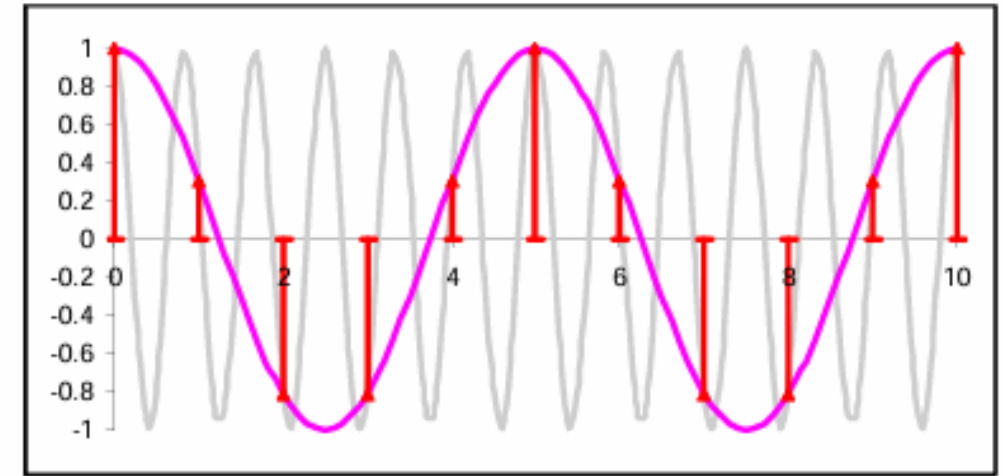
$$X_2(n) = \cos(2\pi [1/10] n)$$

$$X_2(n) = X_1(n)$$

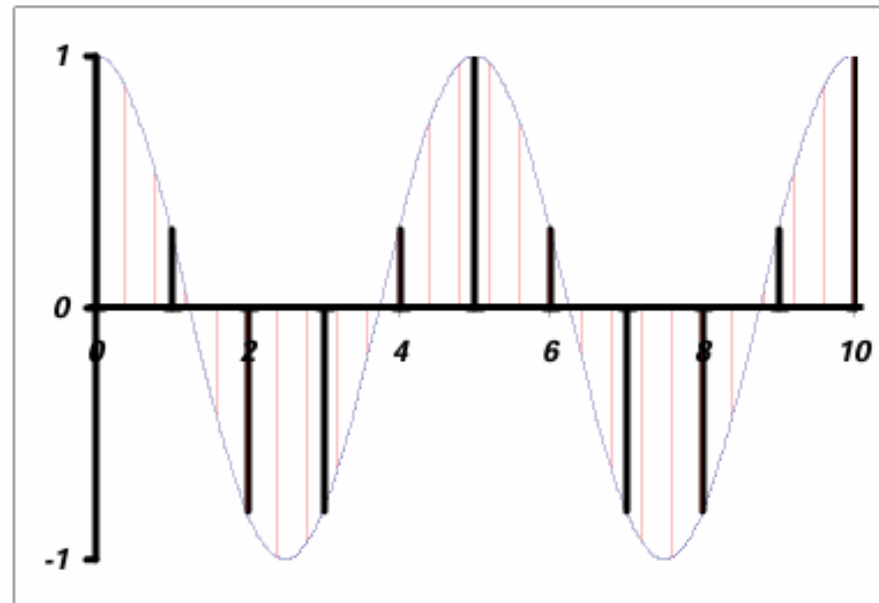
Hence 900 Hz signal will appear as an **alias** of 100 Hz signal



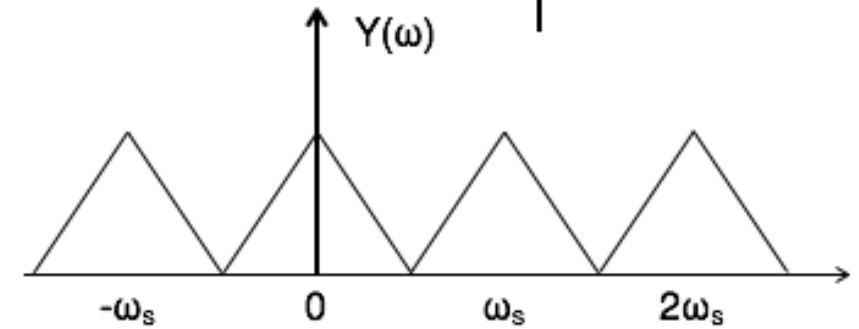
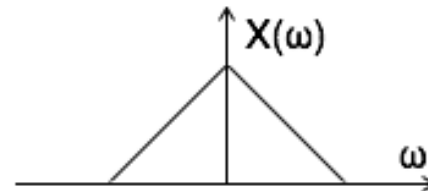
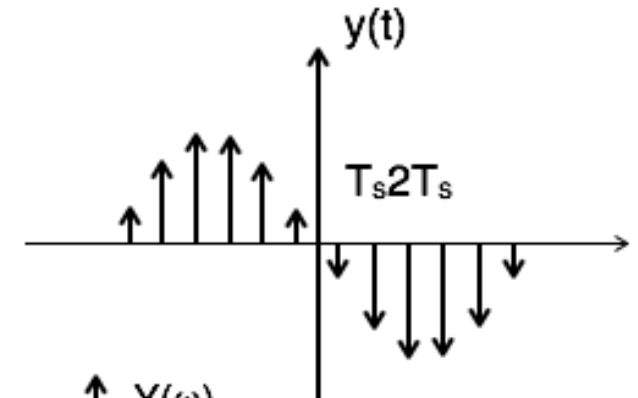
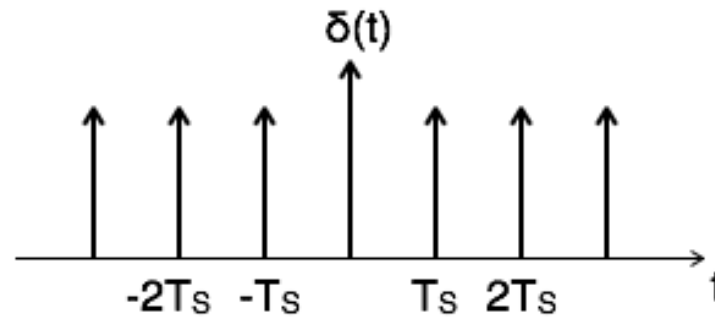
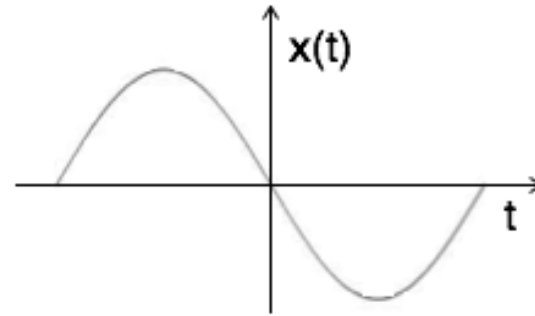
Sampling / Oversampling



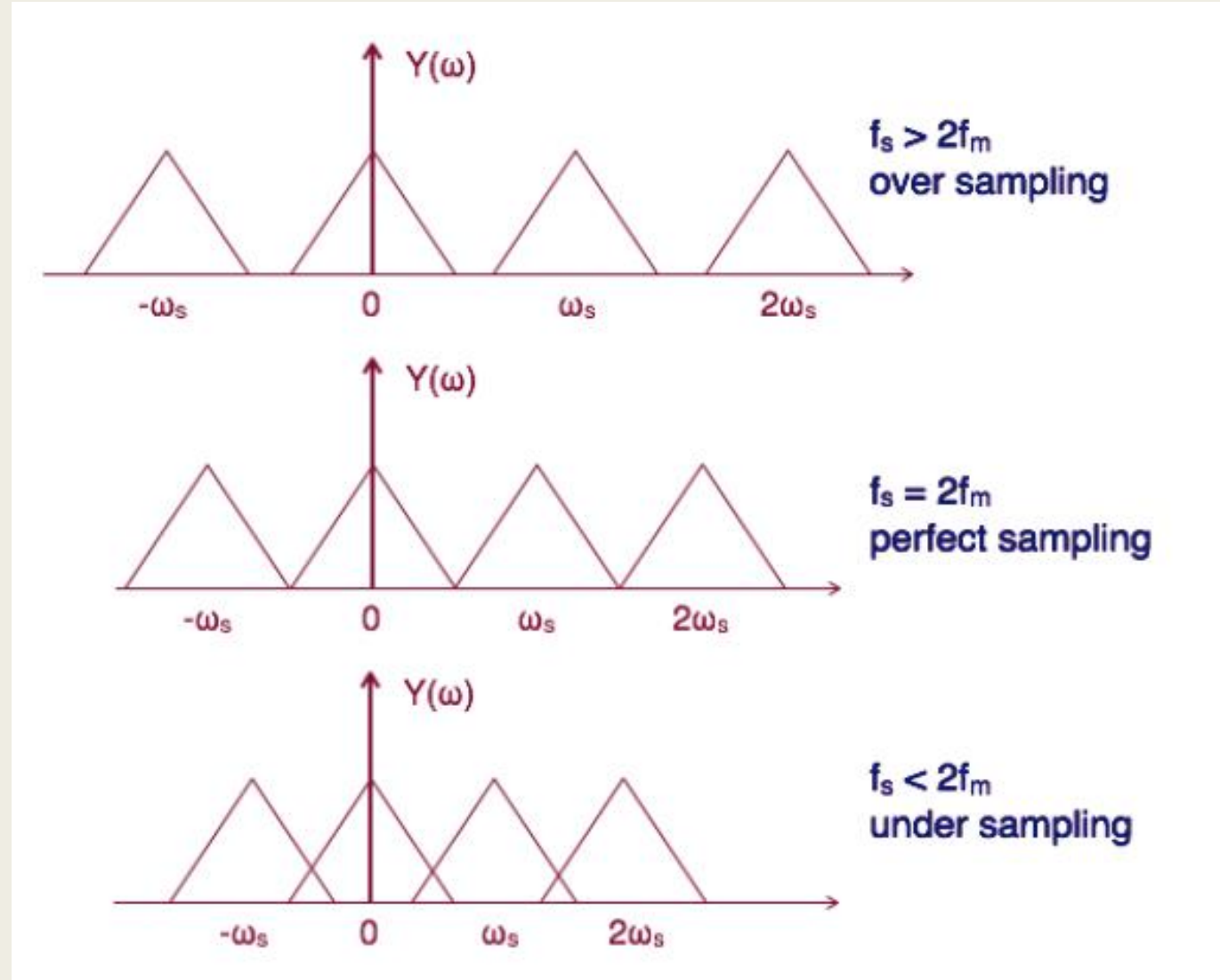
Under sampling



Sampling

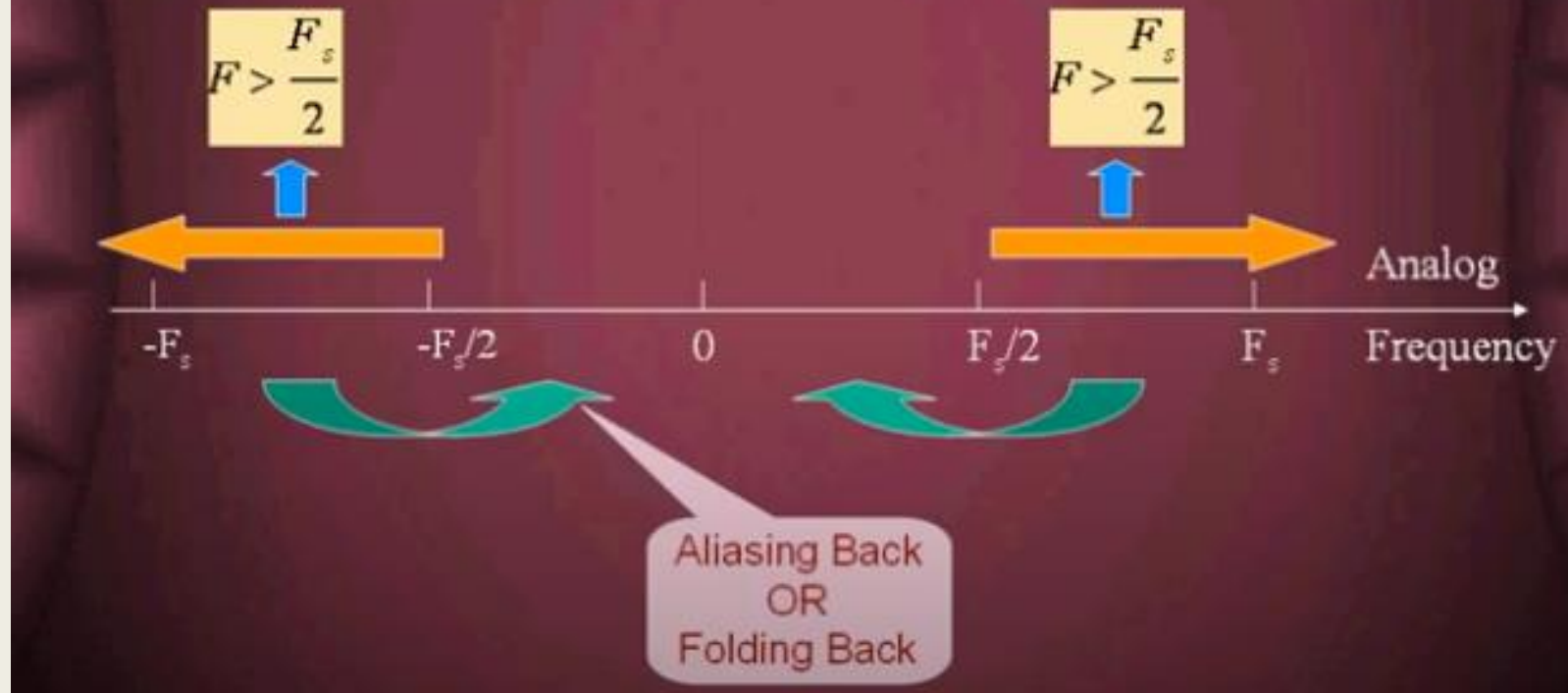


Aliasing



Aliasing

“Transformation of high frequency information into false low frequencies”

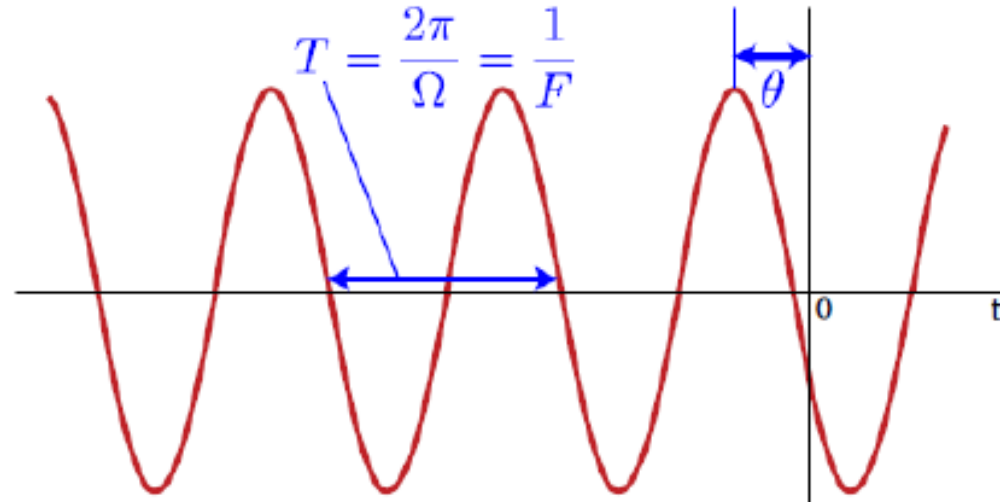


Concept of *Frequency*

CT Sinusoids: What is a “pure frequency” signal?

$$x_a(t) = A \cos(\Omega t + \theta) = A \cos(2\pi F t + \theta) \quad ; t \in \mathbb{R}$$

- Analog signal: $-A < x_a(t) < A$ and $-\infty < t < \infty$
- A = Amplitude (*units depends on nature of signal e.g. for voltage signal, units are volts*)
- Ω = Frequency in *rad/s*
- $\Omega = 2\pi F$
- F = Frequency in *Hz* or *cycles/s*
- θ = Phase in *rad*



Concept of *Frequency*

CT Sinusoids: What is a “pure frequency” signal?

$$x_a(t) = A \cos(\Omega t + \theta) = A \cos(2\pi F t + \theta) \quad ; t \in \mathbb{R}$$

- 1 • For $F \in \mathbb{R}$, $x_a(t)$ is periodic with period $= T_p$. It means there exists a $T_p \in \mathbb{R}^+$ such that $x_a(t) = x_a(t + T_p)$
- 2 • Distinct frequencies result in distinct sinusoids i.e. if $F_1 \neq F_2$, $A \cos(2\pi F_1 t + \theta) \neq A \cos(2\pi F_2 t + \theta)$
- 3 • Increasing frequency results in increase in rate of oscillation of the sinusoid i.e. if $|F_1| < |F_2|$, $A \cos(2\pi F_1 t + \theta)$ has a lower rate of oscillation than $A \cos(2\pi F_2 t + \theta)$

Concept of *Frequency*

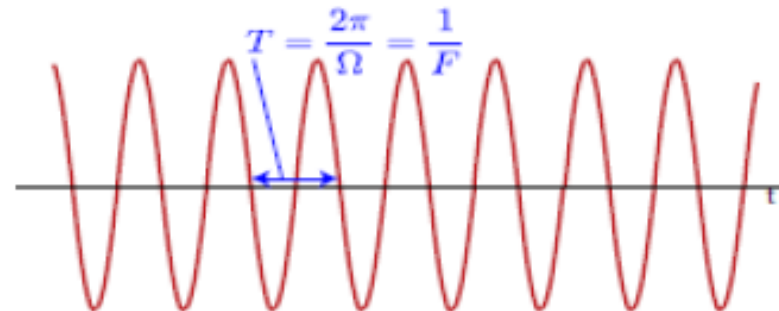
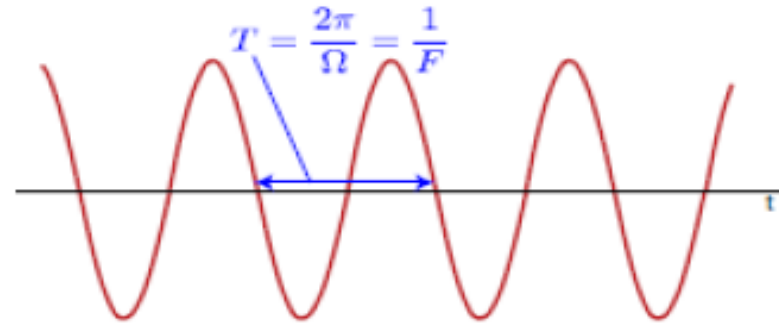
CT Sinusoids: What is a “pure frequency” signal?

$$x_a(t) = A \cos(\Omega t + \theta) = A \cos(2\pi F t + \theta) \quad ; t \in \mathbb{R}$$

$$F = \frac{1}{T}$$

Smaller $F \Rightarrow$ Larger T

Larger $F \Rightarrow$ Smaller T

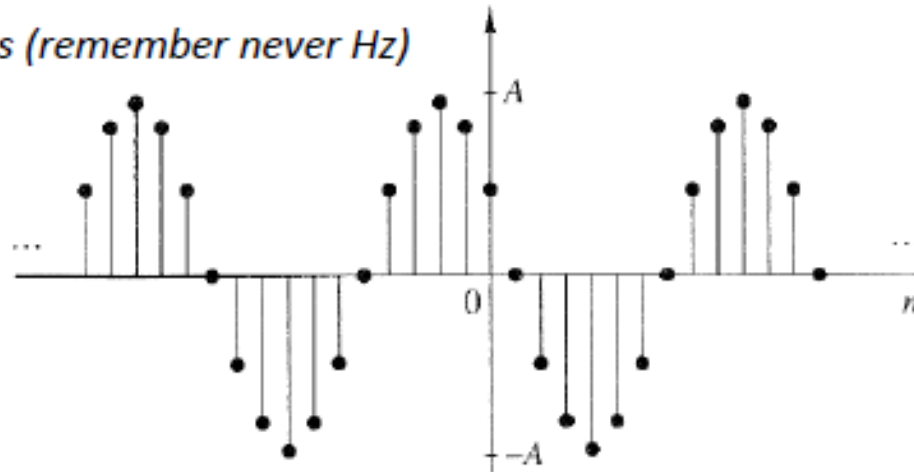


Concept of *Frequency*

DT Sinusoids: What is a “pure frequency” signal?

$$x(n) = A \cos(\omega n + \theta) = A \cos(2\pi f n + \theta) \quad ; n \in \mathbb{Z}$$

- Discrete time signal (not digital !): $-A < x(n) < A$ and $n \in \mathbb{Z}$
- A = Amplitude (*units depends on nature of signal e.g. for voltage signal, units are volts*)
- ω = Frequency in *rad/sample*
- $\omega = 2\pi f$
- f = Frequency in *cycles/samples* (*remember never Hz*)
- θ = Phase in *rad*



Concept of *Frequency*

DT Sinusoids: What is a “pure frequency” signal?

$$x(n) = A \cos(\omega n + \theta) = A \cos(2\pi f n + \theta) \quad ; n \in \mathbb{Z}$$

1

- $x(n)$ is periodic only if its frequency is a RATIONAL NUMBER

Note: rational number is of the form $\frac{k_1}{k_2}$ for $k_1, k_2 \in \mathbb{Z}$

- Periodic sinusoids: $x(n) = 2 \cos\left(\frac{2\pi}{7}n\right) \Rightarrow f = \frac{1}{7}$
- Non-periodic sinusoids: $x(n) = 2 \cos\left(\frac{2}{7}n\right) \Rightarrow f = \frac{1}{7\pi}$

2

- Radian frequencies separated by integer multiple of 2π are identical

OR

cyclic frequencies separated by an integer multiple are identical.

$$\cos\left(\frac{2\pi}{7}n\right) \text{ and } \cos\left(\frac{16\pi}{7}n\right) \text{ are identical as } f_2 - f_1 = 1$$

Concept of *Frequency*

DT Sinusoids: What is a “pure frequency” signal?

$$x(n) = A \cos(\omega n + \theta) = A \cos(2\pi f n + \theta) \quad ; n \in \mathbb{Z}$$

3

- Lowest rate of oscillation is obtained when

$$\omega = 2k\pi \text{ or } f = k \text{ (integer)}$$

- Highest rate of oscillation is obtained when

$$\omega = (2k+1)\pi \text{ or } f = \frac{k+1}{2} \text{ (half integer)}$$

for $k \in \mathbb{Z}$.

Discrete-time Sinusoids

Example i: $\omega = \frac{37}{11}\pi$

$$N = \frac{2\pi k}{\omega} = \frac{2\pi k}{\frac{37}{11}\pi} = \frac{22}{37}k$$

$$N_0 = \frac{22}{37}k = \boxed{22} \text{ for } k = 37; x[n] \text{ is periodic.}$$

Example ii: $\omega = 2$

$$N = \frac{2\pi k}{\omega} = \frac{2\pi k}{2} = \pi k$$

$N \in \mathbb{Z}^+$ does not exist for any $k \in \mathbb{Z}$; $x[n]$ is non-periodic.

Example iii: $\omega = \sqrt{2}\pi$

$$N = \frac{2\pi k}{\omega} = \frac{2\pi k}{\sqrt{2}\pi} = \sqrt{2}k$$

$N \in \mathbb{Z}^+$ does not exist for any $k \in \mathbb{Z}$; $x[n]$ is not periodic.

Discrete-time Sinusoids

$$N = \frac{2\pi k}{\omega}$$
$$\omega = \frac{2\pi k}{N} = 2\pi \frac{k}{N} = \pi \cdot \underbrace{\frac{2k}{N}}_{\text{RATIONAL}}$$

Therefore, a discrete-time sinusoid is periodic if its radian frequency ω is a rational multiple of π .

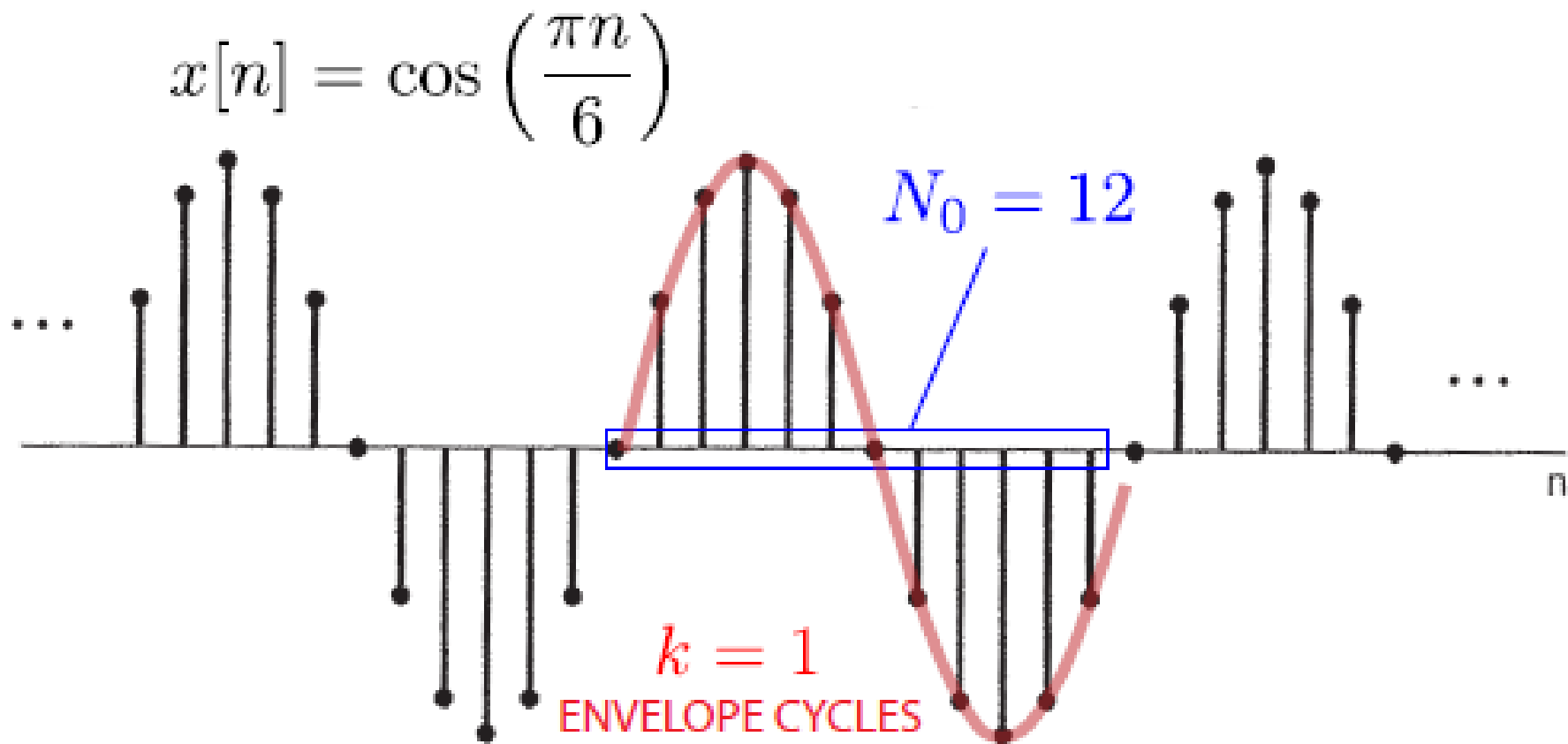
Otherwise, the discrete-time sinusoid is non-periodic.

Example 1: $\omega = \pi/6 = \pi \cdot \boxed{\frac{1}{6}}$

$$x[n] = \cos\left(\frac{\pi n}{6}\right)$$

$$N = \frac{2\pi k}{\omega} = \frac{2\pi k}{\pi \frac{1}{6}} = 12k$$
$$N_0 = 12 \quad \text{for } k = 1$$

The fundamental period is 12 which corresponds to $k = 1$ envelope cycles.



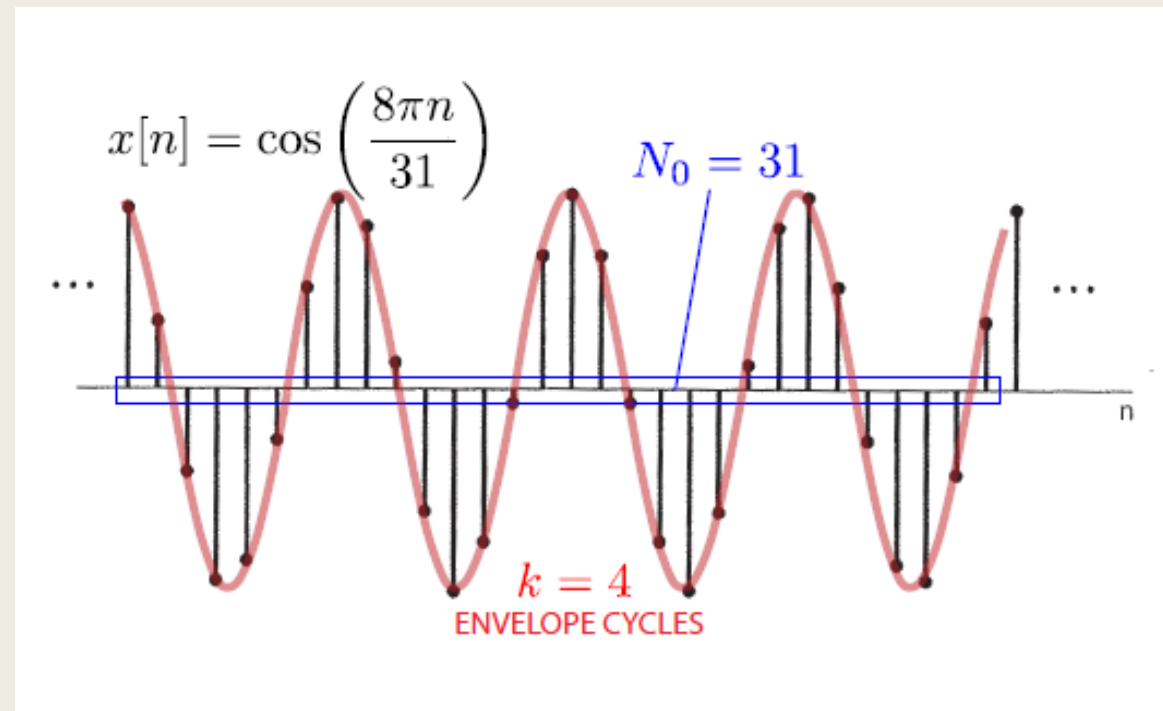
Example 2: $\omega = 8\pi/31 = \pi \cdot \boxed{\frac{8}{31}}$

$$x[n] = \cos\left(\frac{8\pi n}{31}\right)$$

$$N = \frac{2\pi k}{\omega} = \frac{2\pi k}{\pi \frac{8}{31}} = \frac{31}{4}k$$

$$N_0 = 31 \quad \text{for } k = 4$$

The fundamental period is 31 which corresponds to $k = 4$ envelope cycles.

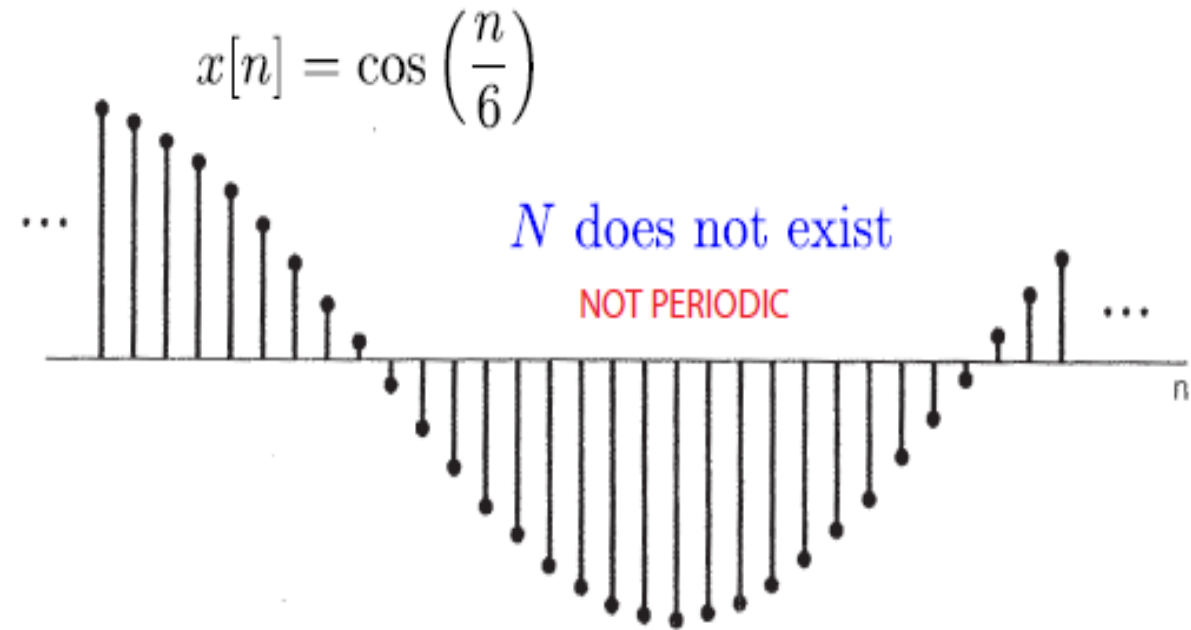


Example 3: $\omega = 1/6 = \pi \cdot \boxed{\frac{1}{6\pi}}$

$$x[n] = \cos\left(\frac{n}{6}\right)$$

$$N = \frac{2\pi k}{\omega} = \frac{2\pi k}{\frac{1}{6}} = 12\pi k$$

$N \in \mathbb{Z}^+$ does not exist for any $k \in \mathbb{Z}$; $x[n]$ is non-periodic.



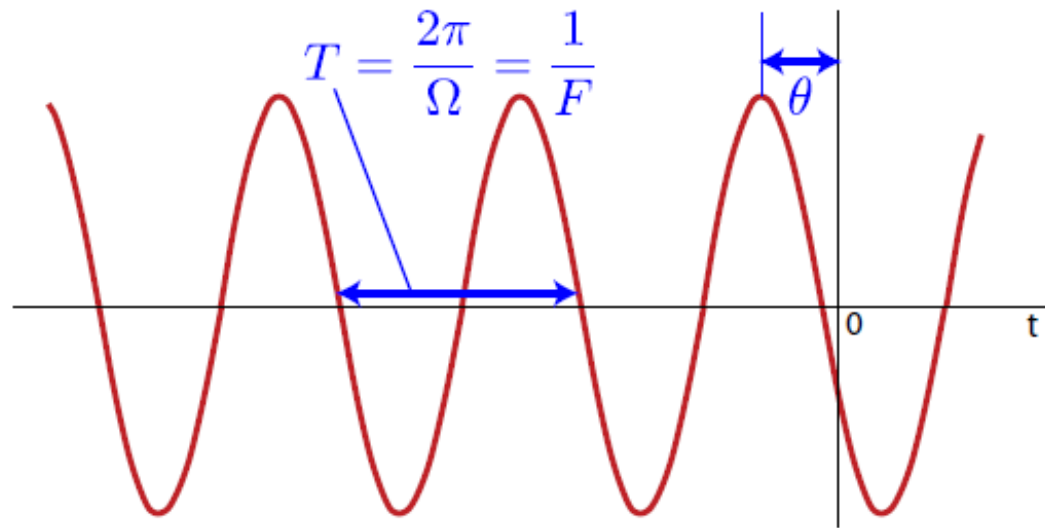
Continuous-Time Sinusoids: Frequency and Rate of Oscillation

$$x(t) = A \cos(\Omega t + \phi)$$

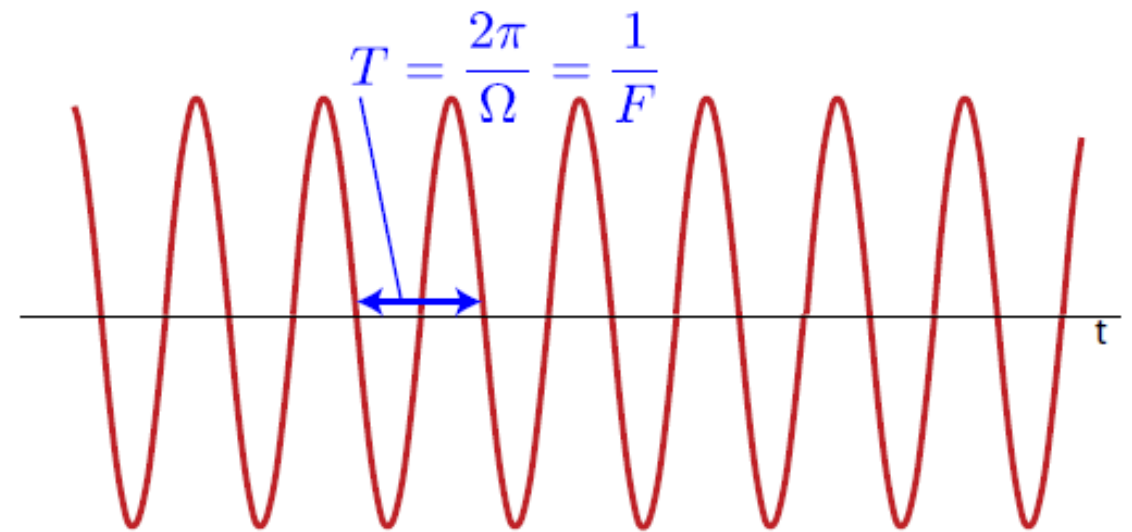
$$T = \frac{2\pi}{\Omega} = \frac{1}{f}$$

Rate of oscillation increases as Ω increases (or T decreases).

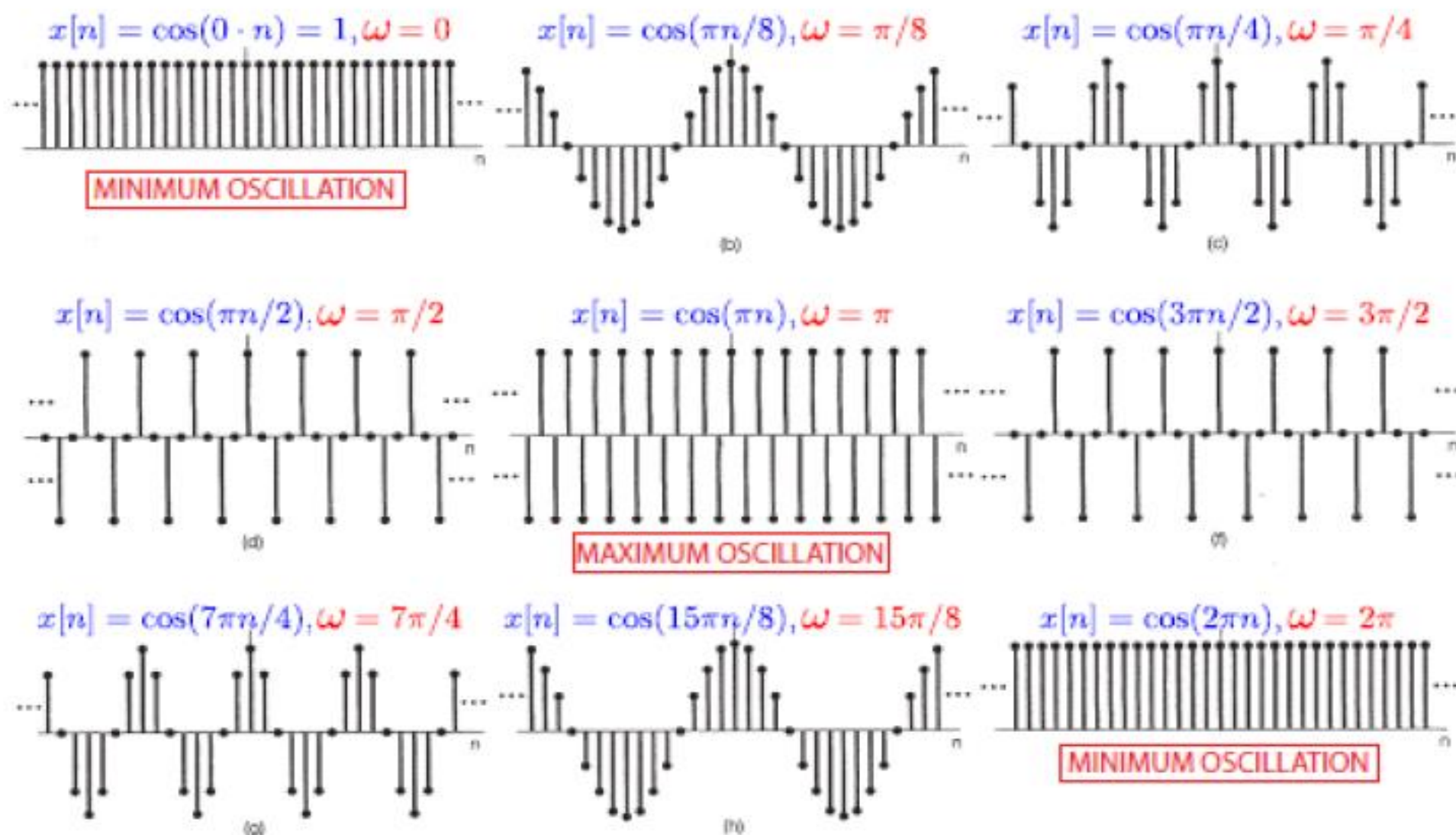
Ω smaller



Ω larger, rate of oscillation higher



Concept of *Frequency*



Complex Exponentials

Complex exponentials play a major role in the analysis of signals and systems, so, we need to review them

$$e^{j\phi} = \cos(\phi) + j \sin(\phi)$$

Euler's Relation

$$\cos(\phi) = \frac{e^{j\phi} + e^{-j\phi}}{2}$$

$$\sin(\phi) = \frac{e^{j\phi} - e^{-j\phi}}{2j}$$

$$\text{where } j \triangleq \sqrt{-1}$$

Complex Exponentials

Complex exponentials play a major role in the analysis of signals and systems, so, we need to review them

Continuous-time: $Ae^{j(\Omega t + \theta)} = Ae^{j(2\pi Ft + \theta)}$

Discrete-time: $Ae^{j(\omega n + \theta)} = Ae^{j(2\pi fn + \theta)}$

Rotating Phasors: Rate of rotation is given by angular frequencies

Complex Exponentials

We will explore **THREE** important characteristics of a Complex Exponential

- **Periodicity**
- **Uniqueness**
- **Fundamental Frequency Range**

Complex Exponentials

Periodicity CT:

$$\begin{aligned}x(t) &= x(t + T), \quad T \in \mathbb{R}^+ \\Ae^{j(2\pi Ft + \theta)} &= Ae^{j(2\pi F(t+T) + \theta)} \\e^{j2\pi Ft} e^{j\theta} &= e^{j2\pi Ft} e^{j2\pi FT} e^{j\theta} \\1 &= e^{j2\pi FT} \\e^{j2\pi k} &= 1 = e^{j2\pi FT}, \quad k \in \mathbb{Z} \\T &= \frac{k}{F} \\\text{Fundamental period} &= T_0 = \frac{1}{|F|}\end{aligned}$$

Complex Exponentials

Periodicity DT:

$$\begin{aligned}x(n) &= x(n + N), \quad N \in \mathbb{R} \\Ae^{j(2\pi fn + \theta)} &= Ae^{j(2\pi f(n+N) + \theta)} \\e^{j2\pi fn} e^{j\theta} &= e^{j2\pi fn} e^{j2\pi fN} e^{j\theta} \\1 &= e^{j2\pi fN} \\e^{j2\pi k} &= 1 = e^{j2\pi fN}, \quad k \in \mathbb{Z} \\f &= \frac{k}{N} \\\text{Fundamental Period: } N_0 &= \frac{k'}{f}, \quad \min |k'| \in \mathbb{Z} \text{ such that } \frac{k'}{f} \in \mathbb{Z}^+\end{aligned}$$

Complex Exponentials

Periodicity DT:

Example 1: $x(n) = \cos\left(\frac{\pi n}{6}\right)$

$$f = \frac{\omega}{2\pi} = \frac{\pi}{2\pi \cdot 6} = \frac{1}{12} \Rightarrow \text{Rational Number}$$

$$N = \frac{k}{f} = 12k$$

Fundamental Period: $N_0 = 12$ for $k = 1$

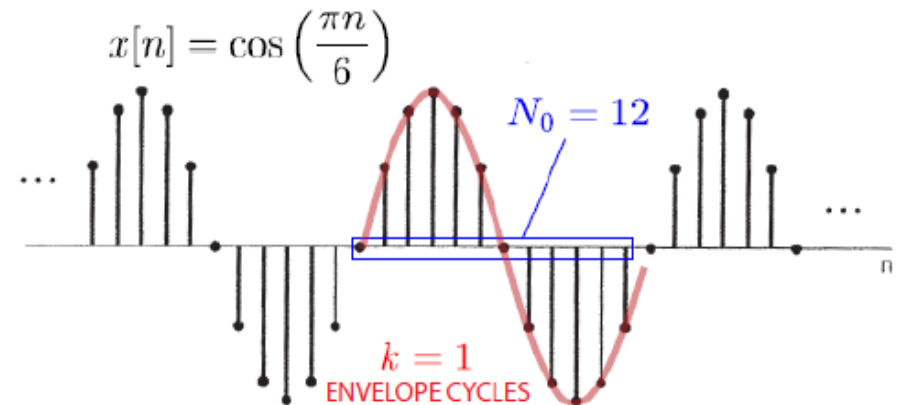
The fundamental period is 12 which corresponds to $k = 1$ envelop cycles

Complex Exponentials

Periodicity DT:

Example 1: $x(n) = \cos\left(\frac{\pi n}{6}\right)$

The fundamental period is 12 which corresponds to $k = 1$ envelop cycles



Complex Exponentials

Periodicity DT:

Example 2: $x(n) = \cos\left(\frac{8\pi n}{31}\right)$

$$f = \frac{\omega}{2\pi} = \frac{8\pi}{2\pi \cdot 31} = \frac{4}{31} \Rightarrow \text{Rational Number}$$

$$N = \frac{k}{f} = \frac{31}{4}k$$

$$N_0 = 31 \quad \text{for } k = 4$$

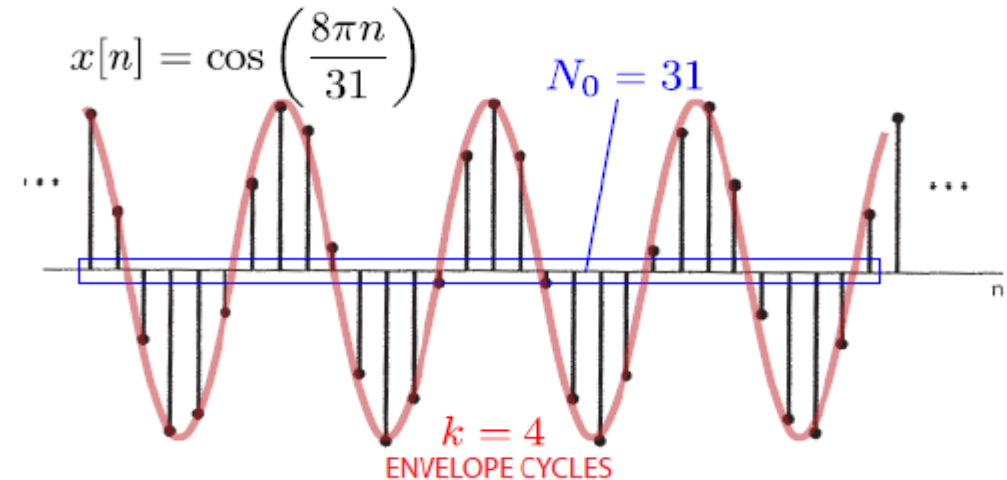
The fundamental period is 31 which corresponds to $k = 4$ envelop cycles

Complex Exponentials

Periodicity DT:

Example 2: $x(n) = \cos\left(\frac{8\pi n}{31}\right)$

The fundamental period is 31 which corresponds to $k = 4$ envelop cycles



Complex Exponentials

Periodicity DT:

Example 3: $x(n) = \cos\left(\frac{n}{6}\right)$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi \cdot 6} = \frac{1}{12\pi} \Rightarrow \text{Irrational Number}$$

$$N = \frac{k}{f} = 12\pi k$$

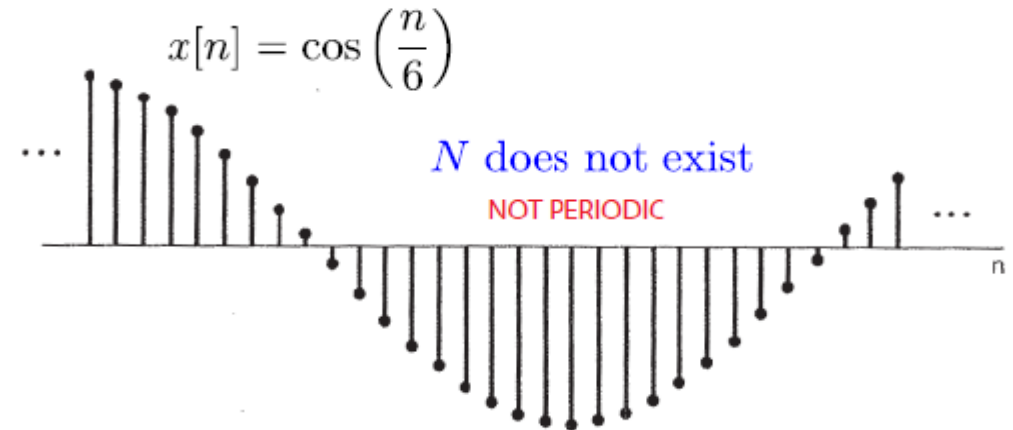
$N \in \mathbb{Z}^+$ does not exist for any $k \in \mathbb{Z}$;

$x(n)$ is non-periodic

Complex Exponentials

Periodicity DT:

Example 3: $x(n) = \cos\left(\frac{n}{6}\right)$



Complex Exponentials

Uniqueness CT:

For $F_1 \neq F_2$,

$$A \cos(2\pi F_1 t + \theta) \neq A \cos(2\pi F_2 t + \theta)$$

So, continuous time sinusoids and complex exponentials having different frequencies are always unique except at some discrete points in time

Complex Exponentials

Uniqueness DT:

For two signals $x_0(n)$ and $x_1(n)$, let $f_1 = f_0 + k$ where $k \in \mathbb{Z}$,

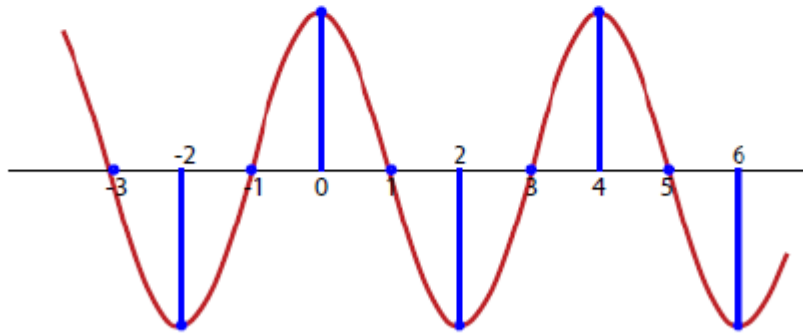
$$\begin{aligned}x_1(n) &= Ae^{j(2\pi f_1 n + \theta)} \\&= Ae^{j(2\pi(f_0 + k)n + \theta)} \\&= Ae^{j(2\pi f_0 n + \theta)} e^{j(2\pi kn)} \\&= x_0(n) \cdot 1 = x_0(n)\end{aligned}$$

Both signals are same ! But HOW ?

Complex Exponentials

Uniqueness DT:

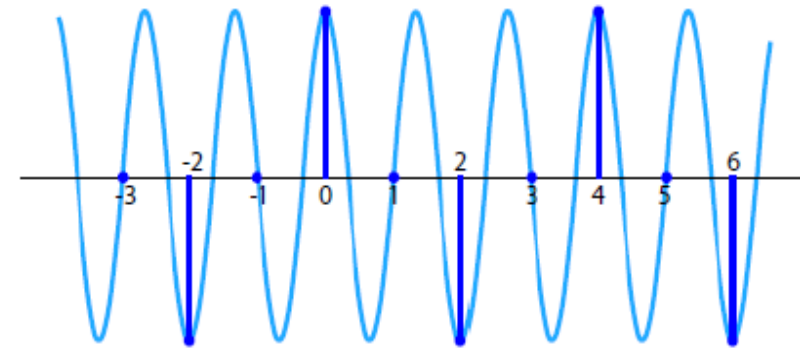
Both signals are same ! **But HOW ?**



Complex Exponentials

Uniqueness DT:

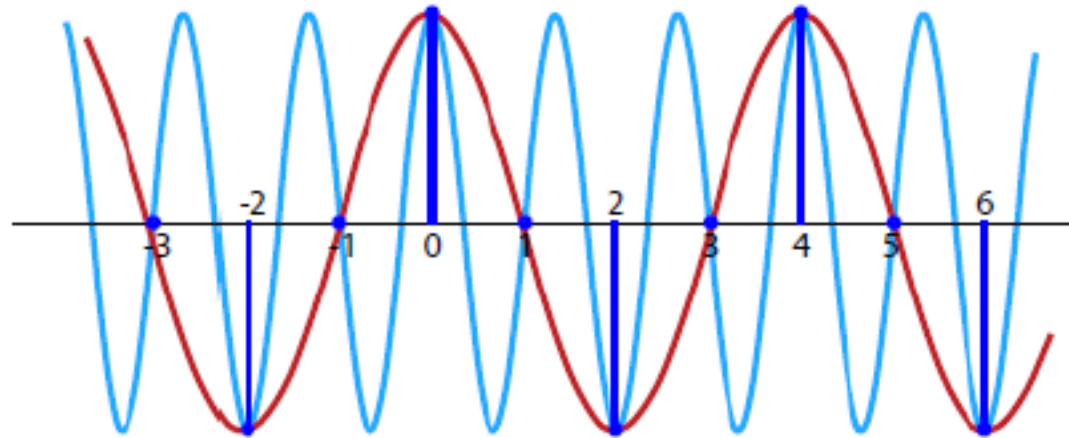
Both signals are same ! **But HOW ?**



Complex Exponentials

Uniqueness DT:

Both signals are same ! **But HOW ?**



Complex Exponentials

Fundamental Frequency Range:

Frequency range of discrete time sinusoids is finite

$$\begin{array}{ccc} 0 \leq \omega \leq 2\pi & \text{OR} & -\pi \leq \omega \leq \pi \\ \downarrow & & \downarrow \\ 0 \leq f \leq 1 & \text{OR} & -\frac{1}{2} \leq f \leq \frac{1}{2} \end{array}$$

Thank You !!!